

We will return at 11:50 am.

Arguments, Rules of Inference

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Learning Outcome

At the end of this lecture, you should be able to:

- Understand what is an argument and what is a valid argument.
- Understand and apply the rules of inference.

Arguments

An **argument** is an assertion that a given set of propositions $P_1, \dots, P_n$ called **premises** (or **hypotheses**), implies a new proposition Q , called the **conclusion**.

Such argument is denoted by $P_1, \dots, P_n \vdash Q$ or

$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline \therefore Q \end{array}$$

An argument is **valid** if (when all the premises $P_1, \dots, P_n$ is true, Q is true.)
In other words, the argument $P_1, \dots, P_n \vdash Q$ is valid if and only if
($P_1 \wedge \dots \wedge P_n \rightarrow Q$ is a tautology.) If an argument is false, it is called a **fallacy**.

To check the validity of an argument, we use the **rules of inference**.

Arguments

If it rains, I drive to work. $\leftarrow P1 : p \rightarrow q$

Example: I didn't drive to work. $\leftarrow P2 : \sim q$

$\therefore$ It did not rain. $\leftarrow Q : \sim p$

Test the validity of the argument above.

Let p : It rains.

q : I drive to work.

Test whether
 $[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p \rightarrow q$
 is a tautology. ✓

Symbolically, the argument becomes $[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$.

p	q	$[(p \rightarrow q) \wedge \sim q]$	$\rightarrow$	$\sim p$
T	T	F	T	F
T	F	F	T	F
F	T	F	T	T
F	F	T	T	T

So, the argument
 is valid.

↑
 tautology.

Rules of Inference

For propositions p, q, r , we have the following rules of inference:

~~**~~ Law of Detachment (Modus Ponens)

$$\frac{\begin{array}{c} p \\ p \rightarrow q \end{array}}{\therefore q}$$

* Modus Tollens

$$\frac{\begin{array}{c} p \rightarrow q \\ \sim q \end{array}}{\therefore \sim p}$$

** Disjunctive Addition

$$\frac{p}{\therefore p \vee q}$$

** Conjunctive Simplification

$$\frac{p \wedge q}{\therefore p}$$

Rules of Inference

*✱ Disjunctive Syllogism

$$\frac{p \vee q \quad \sim p}{\therefore q}$$

$$\frac{p \vee q \quad \sim q}{\therefore p}$$

*✱ Conjunction

$$\frac{p \quad q}{\therefore p \wedge q}$$

$$\frac{p \quad q \quad r}{\therefore p \wedge q \wedge r}$$

*✱ Law of the Syllogism

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

*✱ Resolution

$$\frac{p \vee q \quad \sim p \vee r}{\therefore q \vee r}$$

Rules of Inference

Example: Resolution

$$\frac{p \vee q \quad \sim p \vee r}{\therefore q \vee r}$$

MI: Verify $\#$
 $[(p \vee q) \wedge (\sim p \vee r)] \rightarrow (q \vee r)$
 $\Rightarrow$ a tautology.

u2.

p	q	r	$\sim p$	$p \vee q$	$\sim p \vee r$	$q \vee r$
T	T	T	F	T	T	T
T	T	F	F	T	F	T
T	F	T	F	T	T	T
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	T	F	T	T
F	F	F	T	F	T	F

Critical row
 The corresponding row in the truth table where all the premises are true.

Critical row
 Critical row
 Critical row
 Critical row

Since the conclusion in all the critical rows are true, the argument is valid.

Rules of Inference

M3.

Example: Prove that the following argument is valid.

$$\begin{array}{l} \checkmark \quad p \rightarrow q \leftarrow \text{Premise 1} \\ \checkmark \quad \sim p \rightarrow \sim r \leftarrow \text{Premise 2} \\ \checkmark \quad \quad \quad r \leftarrow \text{Premise 3} \\ \hline \quad \quad \quad \therefore q \leftarrow \text{Conclude.} \end{array}$$

Steps.

- (1) $p \rightarrow q$
- (2) $\sim p \rightarrow \sim r$
- (3) r
- (4) p
- (5) $\therefore q$

Premise 1

Premise 2

Premise 3.

(2), (3), Modus Tollens

(1), (4), Modus Ponens

Note: These steps
are not
generally
unique.

Steps.

(1) $p \rightarrow q$

Premise

(2) $\sim p \rightarrow \sim r$

Premise

(3) r

Premise

(4) $r \rightarrow p$

Contrapositive of (2)

(5) p

(3), (4) modus Ponens.

(6) q

(1), (5) modus Ponens.

Rules of Inference

Example: Use rules of inferences to prove that the following argument is valid.

(If the band could not perform or the refreshments were not delivered on time, then the party would have been cancelled and Amy would have been angry.)
(Refunds would have had to be made unless the party did not cancel.)
(No refunds were made.)
Therefore, (the band could perform.)

P_1
 P_2
 P_3
 Q

Let p : The band could perform.

q : The refreshment were delivered on time.

r : The party was cancelled.

s : Amy was angry.

t : Refund were made.

$$P_1: (\neg p \vee \neg q) \rightarrow (r \wedge s)$$

$$P_2: t \vee \neg r$$

$$P_3: \neg t \quad Q: p$$

Rules of Inference

Argument :

$(\neg p \vee \neg q) \rightarrow (r \wedge s)$	← premise 1
$\neg \neg r$	← premise 2
$\neg t$	← premise 3
$\therefore \underline{\underline{p}}$	← conclusion.

steps

(1) $(\neg p \vee \neg q) \rightarrow (r \wedge s)$

Premise 1

(2) $\neg \neg r$

Premise 2

(3) $\neg t$

Premise 3.

(4) $\neg r$

(2) & (3), Disjunctive Syllogism.

(5) $r \vee \neg s$

(4), Disjunctive addition.

(6) $\neg(r \wedge s)$

(5), De Morgan's Law.

$$(7) \quad \underline{\sim(\sim p \vee \sim q)}$$

$$(8) \quad p \wedge q$$

$$(9) \quad \underline{\therefore p}$$

(1), (6) Modus Tollens.

(7), De Morgan's Law

(8) Conjunctive Simplification.

Rules of Inference

Example: Suppose that

("If you do not attend the classes, you will not pass.") P_1

("If you study hard, you will pass.") P_2

Given that P_3 (Tom passed the course) Did he attend the classes or not? Did he study hard or not?

Let a : Tom attended the classes.

P : Tom passed the course.

s : Tom studied hard.

$$P_1: \sim a \rightarrow \sim P$$

$$(i) Q : a$$

$$P_2: s \rightarrow P$$

$$(ii) Q : s$$

$$P_3: P$$

Rules of Inference

Steps :

$$(1) \quad \underline{\sim a} \rightarrow \underline{\sim p}$$

Premise

$$(2) \quad s \rightarrow p$$

Premise

$$(3) \quad p$$

Premise.

$$(4) \quad \sim(\sim p)$$

(3), Double Negation

$$(5) \quad \sim(\sim a)$$

(1), (4), Modus Tollens

$$(6) \quad \therefore a$$

(5), Double Negation.

So, Tom did attend the class.

a	p	s	$\neg a \rightarrow \neg p$	$s \rightarrow p$
T	<u>T</u>	<u>T</u>	<u>T</u>	<u>T</u>
T	<u>T</u>	<u>F</u>	<u>T</u>	<u>T</u>
T	F	T	T	F
T	F	F	T	T
F	T	T	F	T
F	T	F	F	T
F	F	T	T	F
F	F	F	T	T

So, it is undecidable whether Tom had studied hard.

Rules of Inference

Did you remember the motivation question in the previous class?

Consider the following situation:

Holmes announces, "Watson, I have uncovered the following facts:

- If Mrs Smith is lying then Moriarty has not escaped. $s \rightarrow \sim e$
- Either Moriarty is dead or he is really Jones. $d \vee j$
- If Moriarty is really Jones then he has escaped. $j \rightarrow e$
- Mrs Smith is lying. s

"Good Lord Holmes," replied Dr Watson, "what can you make of all this?"

"Elementary my dear Watson, Moriarty is dead."

Is Holmes correct?

Rules of Inference

Let s : Mrs. Smith is lying
 e : Moriarty has escaped.
 d : Moriarty is dead
 j : Moriarty is really Jones.

steps:

- | | | | |
|-----|------------------------|--------------------------|-------|
| (1) | $s \rightarrow \neg e$ | Premise | |
| (2) | $d \vee j$ | Premise | |
| | | Premise | d ? |
| (3) | $j \rightarrow e$ | Premise. | |
| (4) | s | | |
| (5) | $\neg e$ | (1), (4), Modus Ponens | |
| (6) | $\neg j$ | (3), (5), Modus Tollens. | |

(1) $\therefore d$

(2), (6), Disjunctive Syllogism.

So, Holmes is correct, Moriarty is dead.

* Tutorial 1 & Tutorial 2 Q1-Q7.